

RESEARCH INSTRUMENTS AND SURVEY EXAMPLES

A readable guide for choosing and creating questionnaires, Likert scales, interview guides, observation checklists, experimental tools, and document analysis guides

Purpose of this PDF: This guide helps learners understand what instrument to use for quantitative and qualitative research. Each instrument includes when to use it, what data it produces, how to analyze it, and a clear sample.

Research Type	Best Instrument Examples	Usual Output
Quantitative	Likert-scale questionnaire, structured questionnaire, pre-test and post-test, checklist with counts	Frequency, percentage, weighted mean, standard deviation, t-test, graph
Qualitative	Semi-structured interview guide, focus group guide, observation notes, document analysis guide	Codes, categories, themes, descriptions, participant insights
Can be Both	Observation checklist and document analysis guide	Counts for quantitative or themes for qualitative

Quick rule: If the Statement of the Problem asks for a level, frequency, relationship, difference, or effect, the study is usually quantitative. If it asks for experiences, challenges, perceptions, meanings, or coping strategies, the study is usually qualitative.

1. Instrument Selection Guide

The instrument must match the Statement of the Problem. A wrong instrument can make Chapter 4 weak because the gathered data will not answer the research questions.

Research Question	Recommended Instrument	Research Type	How to Analyze
What is the level of usability of the system?	Likert-scale questionnaire	Quantitative	Weighted mean, standard deviation, rank
What is the profile of the respondents?	Structured questionnaire	Quantitative	Frequency and percentage
Is there a significant difference between students and faculty?	Questionnaire with grouped responses	Quantitative	t-test or adviser-approved test
What are the experiences of students in requesting consultation?	Semi-structured interview guide	Qualitative	Thematic analysis
What behavior is observed while users operate the system?	Observation checklist	Can be both	Counts or observation themes
What is the effect of a learning module on test scores?	Pre-test and post-test	Quantitative experimental	Mean, gain score, paired t-test
What do existing records show about consultation requests?	Document analysis guide	Can be both	Counts, patterns, themes

Do: Match every SOP question with a data source and instrument. Do not: create questions that are unrelated to the research problem.

2. Likert-Scale Questionnaire

Used for: Quantitative research. A Likert-scale questionnaire is used when the researcher wants to measure the level of agreement, satisfaction, usability, acceptability, awareness, or effectiveness.

Example SOP: What is the level of usability of the web-based consultation system in terms of efficiency, effectiveness, learnability, and satisfaction?

Scale	Range	Verbal Interpretation
5	4.21 - 5.00	Strongly Agree / Very High
4	3.41 - 4.20	Agree / High
3	2.61 - 3.40	Neutral / Moderate
2	1.81 - 2.60	Disagree / Low
1	1.00 - 1.80	Strongly Disagree / Very Low

Sample items:

Indicator	Questionnaire Statement	Rating
Efficiency	The system allows users to complete consultation booking quickly.	☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1
Effectiveness	The system helps users accomplish consultation-related tasks correctly.	☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1
Learnability	The system is easy to understand even for first-time users.	☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1
Satisfaction	The user is satisfied with the overall consultation process.	☐ 5 ☐ 4 ☐ 3 ☐ 2 ☐ 1

How to analyze: Compute the weighted mean for each indicator. Then rank the indicators and interpret the results using the scale range.

3. Structured Questionnaire

Used for: Quantitative research. A structured questionnaire collects fixed answers such as profile, categories, yes/no responses, and multiple-choice responses.

Question	Choices / Answer Format
1. What is your user role?	☐ Student ☐ Faculty Member
2. What is your year level?	☐ 1st Year ☐ 2nd Year ☐ 3rd Year ☐ 4th Year ☐ Not Applicable
3. Have you used the consultation system?	☐ Yes ☐ No
4. How often do you request academic consultation?	☐ Never ☐ Rarely ☐ Sometimes ☐ Often ☐ Always
5. What device did you use to access the system?	☐ Phone ☐ Tablet ☐ Laptop/Desktop

How to analyze: Use frequency and percentage. This is commonly used for respondent profile tables in Chapter 4.

4. Semi-Structured Interview Guide

Used for: Qualitative research. An interview guide is used when the study aims to understand lived experiences, challenges, perceptions, opinions, and coping strategies.

Example SOP: What are the lived experiences of college students in requesting academic consultation?

Main Question	Follow-up Probe	Purpose
Can you describe your experience when requesting academic consultation?	What happened before, during, and after the request?	Explore lived experience
What challenges did you encounter?	Can you describe a specific situation?	Identify problems
How did you handle these challenges?	Who helped you or what steps did you take?	Identify coping strategies
What support would make the process easier?	What feature, policy, or process would help?	Generate recommendations

How to analyze: Record or take notes with consent, transcribe the answers, code meaningful statements, group codes into categories, and form themes.

5. Observation Checklist

Used for: Quantitative or qualitative research. Use an observation checklist when the researcher needs to watch and record actual behavior, system use, process flow, compliance, or performance.

Observed Item	Yes	No	Observation Notes
The user can find the consultation booking button without assistance.			
The user can select an available schedule.			
The user receives confirmation after booking.			
The user completes the process without system error.			
The user asks for help while using the system.			

Quantitative use: Count Yes/No responses. Qualitative use: Analyze observation notes for repeated patterns or problems.

6. Experimental Instrument: Pre-Test and Post-Test

Used for: Quantitative experimental research. Use this when the study measures the effect of an intervention, module, strategy, application, or treatment.

Example SOP: Is there a significant difference between the pre-test and post-test scores of students after using the learning module?

Participant	Pre-Test Score	Post-Test Score	Gain Score	Remarks
P1				
P2				
P3				
P4				
P5				

How to analyze: Compare pre-test and post-test scores. Possible tools include mean, gain score, paired t-test, or another adviser-approved statistical test.

7. Document Analysis Guide

Used for: Quantitative or qualitative research. Use this when the study reviews records, reports, policies, system logs, transaction history, or existing documents.

Document / Record	Data to Extract	Purpose
Consultation logs	Number of requests, completed sessions, cancelled sessions	Measure consultation activity
Student feedback forms	Common comments or issues	Identify improvement areas
System logs	Login activity, booking timestamps, errors	Verify system usage
School policy documents	Consultation rules and procedures	Understand official process

Quantitative use: Count records or categories. Qualitative use: Code documents and identify themes or patterns.

8. Focus Group Discussion Guide

Used for: Qualitative research. A focus group discussion gathers insights from a group of participants. It is useful when the researcher wants to understand shared opinions, group experiences, or common recommendations.

FGD Question	Purpose
What common difficulties do students experience when requesting consultation?	Identify shared challenges
What part of the consultation process needs improvement?	Generate group recommendations
How can the school make consultation more accessible?	Identify support needs
What system features would be helpful for students and faculty?	Collect improvement ideas

How to analyze: Summarize group responses, code repeated ideas, and develop themes. The researcher must manage the discussion so that one participant does not dominate.

9. Final Instrument Checklist

Check	Question
Alignment	Does the instrument answer the Statement of the Problem?
Clarity	Can respondents or participants understand the questions easily?
Relevance	Is every item connected to the study?
Ethics	Does the instrument avoid private or harmful questions unless necessary and approved?
Validation	Was the instrument checked by experts or the adviser?
Analysis Plan	Is there a clear way to analyze the responses?

Final reminder: The research instrument is not just a form. It is the bridge between Chapter 1 questions and Chapter 4 findings.